

Welcome to AWS DevOps Assignment - CloudFormation Edition

```
PS C:\Users\den\Projects\aws-devops-assignment> aws cloudformation describe-stacks
--stack-name devops-assignment-stack --query "Stacks[0].Outputs" --output json
[
    {
        "OutputKey": "WebsiteURL",
        "OutputValue": "http://18.224.165.187",
        "Description": "Public URL for the EC2 web server"
    },
    {
        "OutputKey": "S3BucketName",
        "OutputValue": "aws-devops-assignment-cf-631237958452",
        "Description": "Created S3 Bucket Name"
    }
]

PS C:\Users\den\Projects\aws-devops-assignment>
```

```
PS C:\Users\den\Projects\aws-devops-assignment> aws cloudformation validate-template
--template-body file://cloudformation/infrastructure.yaml
{
    "Parameters": [
        {
```

```
        "ParameterKey": "EnvironmentName",
        "DefaultValue": "dev",
        "NoEcho": false,
        "Description": "Environment name for tagging"
      }
  ],
  "Description": "AWS DevOps Assignment - CloudFormation Infrastructure Stack - Updated"
}

PS C:\Users\den\Projects\aws-devops-assignment> aws cloudformation describe-stacks
--stack-name devops-assignment-stack --query "Stacks[0].StackStatus"
"UPDATE_COMPLETE"

PS C:\Users\den\Projects\aws-devops-assignment> aws cloudformation describe-stacks
--stack-name devops-assignment-stack --query "Stacks[0].StackStatus"
"UPDATE_COMPLETE"

PS C:\Users\den\Projects\aws-devops-assignment> aws cloudformation describe-stacks
--stack-name devops-assignment-stack --query "Stacks[0].Outputs" --output json
[
  {
    "OutputKey": "WebsiteURL",
    "OutputValue": "http://18.217.161.168",
    "Description": "Public URL for the EC2 web server"
  },
  {
    "OutputKey": "S3BucketName",
    "OutputValue": "aws-devops-assignment-cf-631237958452",
    "Description": "Created S3 Bucket Name"
  }
]

PS C:\Users\den\Projects\aws-devops-assignment>
```


```
AWSTemplateFormatVersion: '2010-09-09'
Description: "AWS DevOps Assignment - CloudFormation Infrastructure Stack
- Updated"

Parameters:
  EnvironmentName:
    Type: String
```

```yaml
    Default: 'dev'
    Description: 'Environment name for tagging'

Resources:
  # 1. Virtual Private Cloud (VPC)
  MyVPC:
    Type: AWS::EC2::VPC
    Properties:
      CidrBlock: 10.0.0.0/16
      EnableDnsHostnames: true
      EnableDnsSupport: true
      Tags:
        - Key: Name
          Value: !Sub '${EnvironmentName}-vpc'
        - Key: Owner
          Value: 'den'
        - Key: Project
          Value: 'aws-devops-assignment'
        - Key: Environment
          Value: !Ref EnvironmentName

  # 2. Subnet
  MySubnet:
    Type: AWS::EC2::Subnet
    Properties:
      VpcId: !Ref MyVPC
      CidrBlock: 10.0.1.0/24
      MapPublicIpOnLaunch: true
      Tags:
        - Key: Name
          Value: !Sub '${EnvironmentName}-subnet'
        - Key: Owner
          Value: 'den'
        - Key: Project
          Value: 'aws-devops-assignment'
        - Key: Environment
          Value: !Ref EnvironmentName

  # Internet Gateway & Route Table (Required for public EC2 connectivity)
  InternetGateway:
```

```
  Type: AWS::EC2::InternetGateway
  Properties:
    Tags:
      - Key: Name
        Value: !Sub '${EnvironmentName}-igw'

GatewayAttachment:
  Type: AWS::EC2::VPCGatewayAttachment
  Properties:
    VpcId: !Ref MyVPC
    InternetGatewayId: !Ref InternetGateway

PublicRouteTable:
  Type: AWS::EC2::RouteTable
  Properties:
    VpcId: !Ref MyVPC

PublicRoute:
  Type: AWS::EC2::Route
  DependsOn: GatewayAttachment
  Properties:
    RouteTableId: !Ref PublicRouteTable
    DestinationCidrBlock: 0.0.0.0/0
    GatewayId: !Ref InternetGateway

SubnetRouteTableAssociation:
  Type: AWS::EC2::SubnetRouteTableAssociation
  Properties:
    SubnetId: !Ref MySubnet
    RouteTableId: !Ref PublicRouteTable

# 3. Security Group
WebServerSecurityGroup:
  Type: AWS::EC2::SecurityGroup
  Properties:
    GroupDescription: Enable HTTP and SSH access
    VpcId: !Ref MyVPC
    SecurityGroupIngress:
      - IpProtocol: tcp
        FromPort: 80
```

```
        ToPort: 80
        CidrIp: 0.0.0.0/0
        Description: Allow HTTP traffic
      - IpProtocol: tcp
        FromPort: 22
        ToPort: 22
        CidrIp: 0.0.0.0/0
        Description: Allow SSH access (restrict in production)
    Tags:
      - Key: Name
        Value: !Sub '${EnvironmentName}-sg'
      - Key: Owner
        Value: 'den'
      - Key: Project
        Value: 'aws-devops-assignment'

# 4. S3 Bucket
MyS3Bucket:
  Type: AWS::S3::Bucket
  Properties:
    BucketName: !Sub 'aws-devops-assignment-cf-${AWS::AccountId}'
    VersioningConfiguration:
      Status: Enabled
    Tags:
      - Key: Owner
        Value: 'den'
      - Key: Project
        Value: 'aws-devops-assignment'
      - Key: Environment
        Value: !Ref EnvironmentName

# 5. EC2 Instance
WebServerInstance:
  Type: AWS::EC2::Instance
  Properties:
    ImageId: ami-0c55b159cbfafe1f0 # Amazon Linux 2023 AMI for us-east-2
    InstanceType: t3.micro
    SubnetId: !Ref MySubnet
    SecurityGroupIds:
      - !Ref WebServerSecurityGroup
```

```
      UserData:
        Fn::Base64: !Sub |
          #!/bin/bash
          yum update -y
          yum install -y httpd
          systemctl start httpd
          systemctl enable httpd
          echo "<h1>Welcome to AWS DevOps Assignment - CloudFormation
Edition</h1>" > /var/www/html/index.html
      Tags:
        - Key: Name
          Value: !Sub '${EnvironmentName}-web-server'
        - Key: Owner
          Value: 'den'
        - Key: Project
          Value: 'aws-devops-assignment'

Outputs:
  WebsiteURL:
    Description: Public URL for the EC2 web server
    Value: !Sub 'http://${WebServerInstance.PublicIp}'
  S3BucketName:
    Description: Created S3 Bucket Name
    Value: !Ref MyS3Bucket
```